NOVA SCHOOL OF SCIENCE & TECHNOLOGY

Interação Pessoa-Máquina

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SmartGlycoAI

Stage 3: 1st protoype



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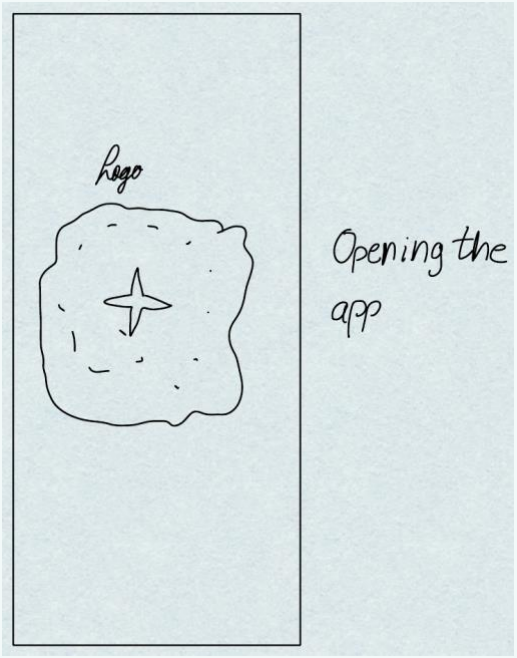
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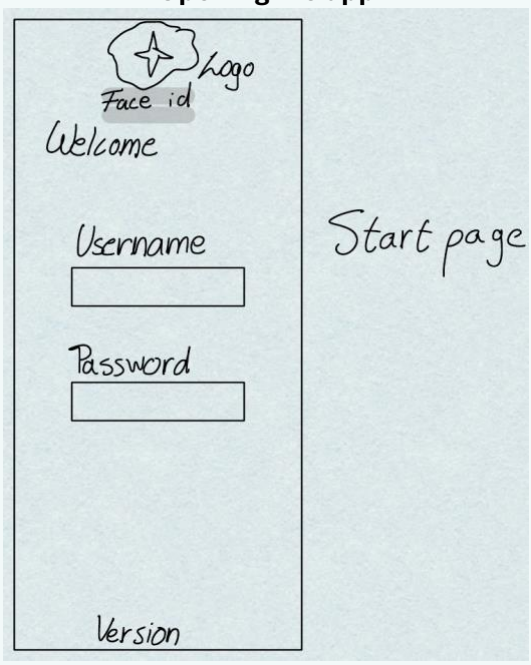
1. Sketches

Objective: Design of the first low/medium fidelity interface prototype, scenario testing preparation, and usability observations.

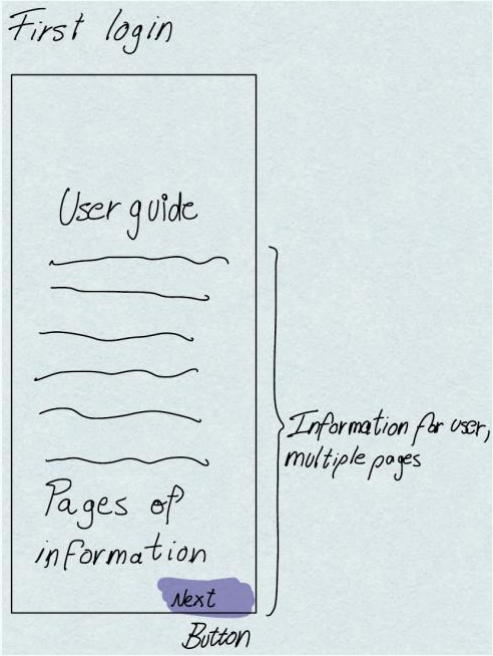
Our first approach to the design: hand-drawn sketches of the main screens, with notes explaining the purpose of each element. They show how the interface looks at the key points of the interaction: opening the app, logging in, the user guide on first login, the Home page, the meal analysis after taking a photo, the Diary and the Profile.



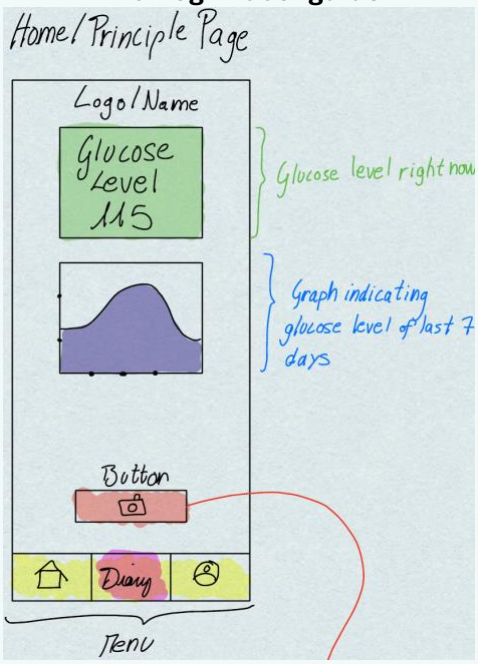
Opening the app



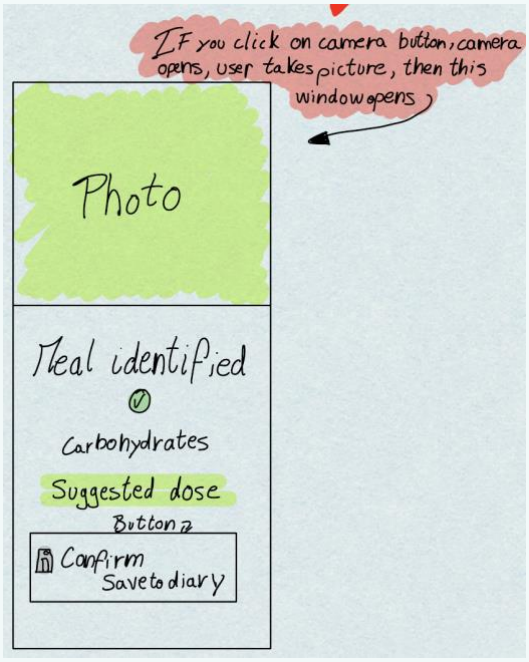
Start page (login)



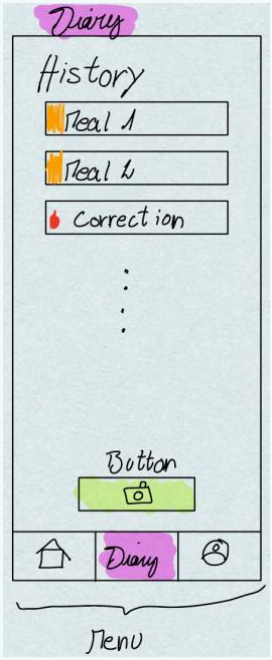
First login: user guide



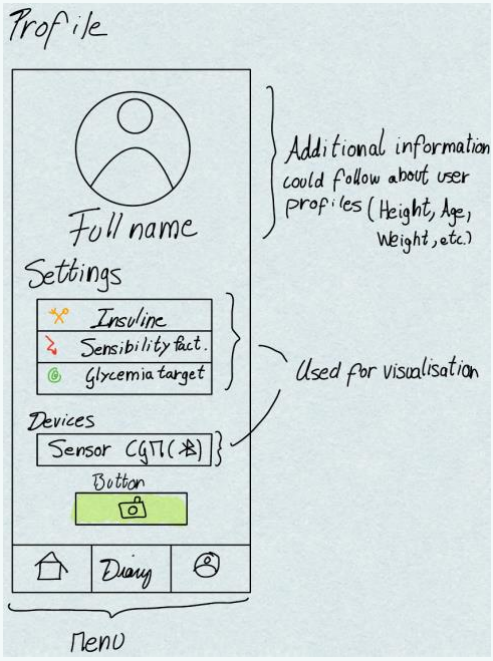
Home / principal page



Meal analysis after the photo



Diary (history)

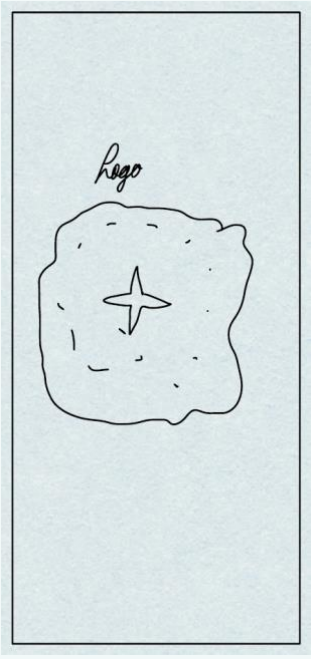


Profile and settings

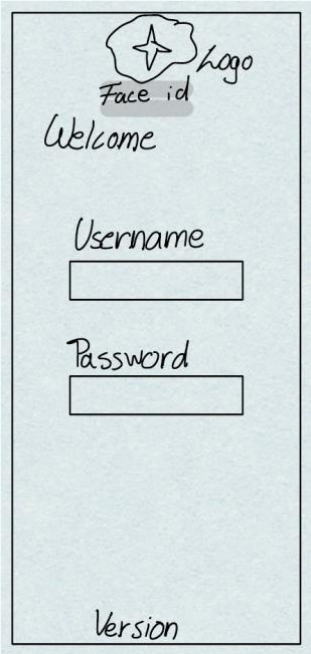
2. Prototype photos

Prototype Interface Photos

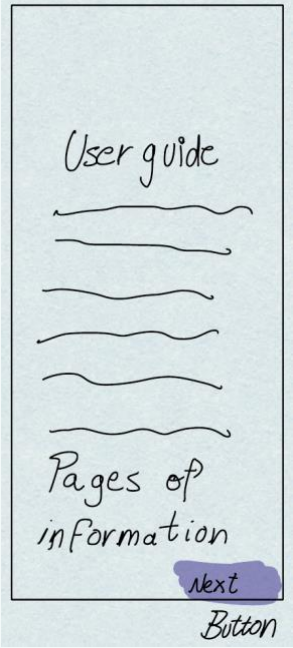
Hand-drawn low-fidelity screens used during the paper prototype evaluation sessions.



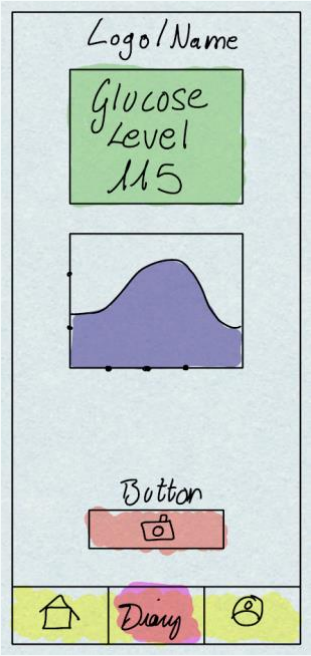
Screen 01 - Opening the App



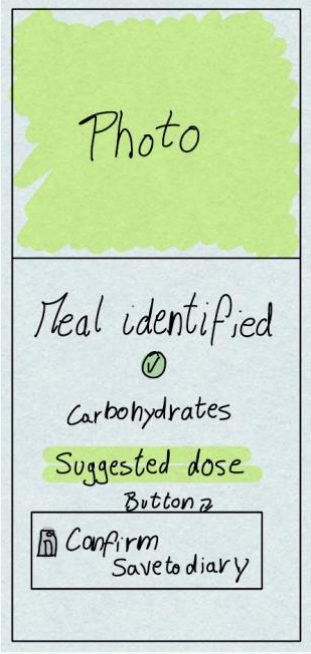
Screen 02 - Prototype Sketch 2 (start page / login)



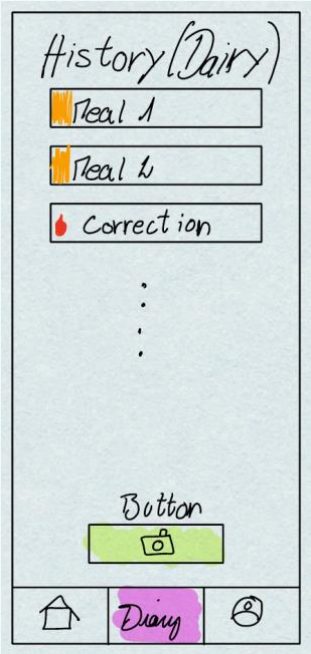
Screen 03 - Prototype Sketch 3 (user guide)



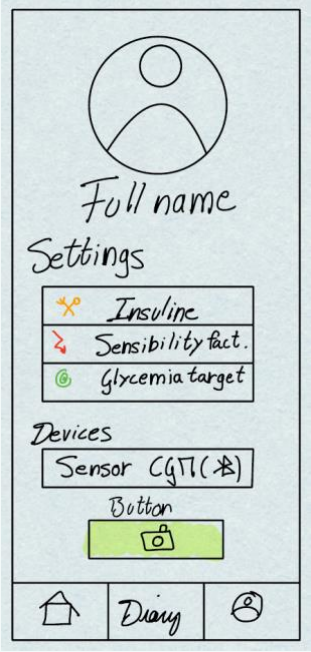
Screen 04 - Prototype Sketch 4 (Home)



Screen 05 - Prototype Sketch 5 (meal analysis)



Screen 06 - Prototype Sketch 6 (Diary)

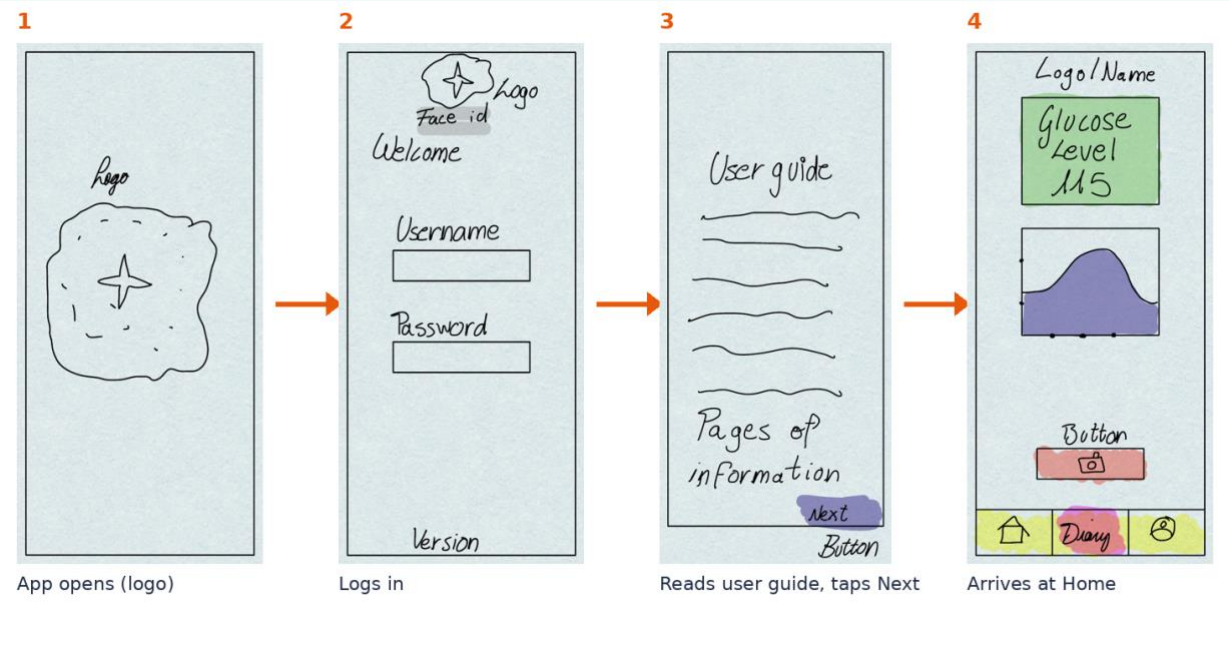


Screen 07 - Prototype Sketch 7 (Profile)

3. Storyboards

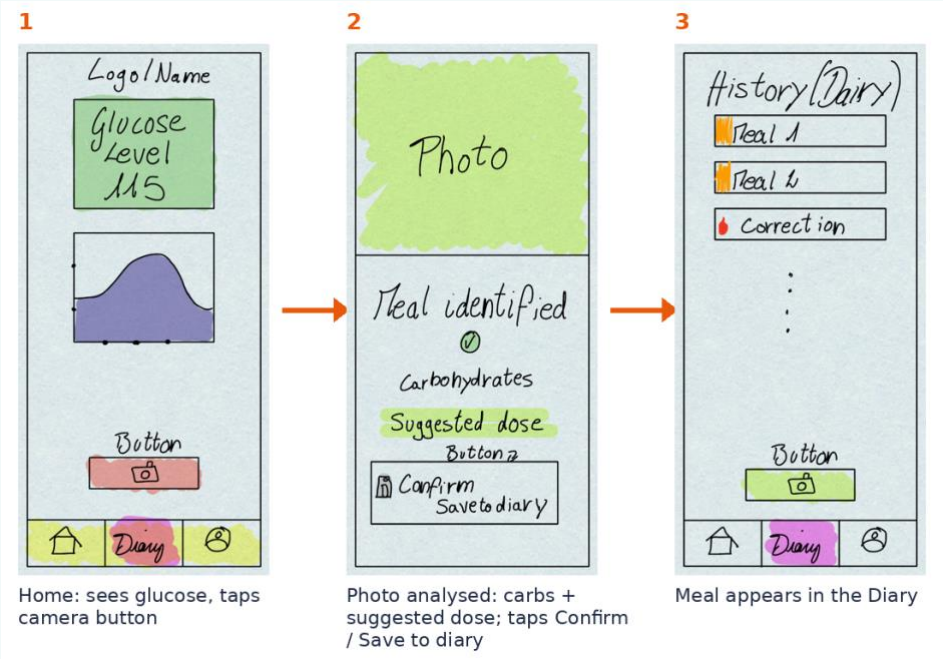
The storyboards below show the sequence of screens a user goes through to perform the most relevant tasks with the paper prototype.

Storyboard A - First use of the app



The user opens the app, logs in, reads the user guide and arrives at the Home page.

Storyboard B - Scenario 1: meal at a restaurant (Task 1)



From the Home page, the user taps the camera button and takes a photo. The app shows the identified meal, the carbohydrates and the suggested dose; the user confirms and the meal is saved in the Diary.

Storyboard C - Scenario 2: correcting the diary (Task 4)



Home: after handling the alert, taps Diary

Diary: finds the lunch entry and opens it to edit the carb

After handling the low-glucose alert, the user opens the Diary from the menu, finds the lunch entry and opens it to change the carbohydrates.

4. Briefing

Testing Preparation

User Briefing (Script Read to Participants)

"Welcome and thank you for participating! Managing Type 1 Diabetes requires constantly balancing food intake, physical activity, and insulin. SmartGlycoAI is an application designed to relieve this burden. It estimates the carbohydrates in your meals, tracks your glucose via a sensor, and suggests the correct insulin dose so you don't have to do the math manually. During this test, we want you to interact with the interface naturally to complete a few tasks. Please 'think aloud' as you go, so we understand your thought process. Remember, we are testing the interface, not you!"

Evaluation Roles

The Computer	Responsible for manipulating the paper prototype, changing screens, and reacting to the user's touch inputs without speaking.
The Facilitator	Reads the briefing, presents the scenarios, and encourages the user to think aloud throughout the evaluation session.
The Observers	Take detailed notes on usability problems, user hesitation, and navigation errors during each task.

Roles were swapped after every test so that each group member played each role.

5. Scenarios

The 3 Scenarios (From Stage 2)

SCENARIO 1 • PATIENT Task 1

Meal Time at a Restaurant

João is at a restaurant and is unsure about the carbohydrate count of his pasta dish. Glancing at his Home screen, he views his stable 6-hour glucose trend graph. He opens SmartGlycoAI's meal logging module, takes a photo of his plate, and the app estimates 65g of carbs. Based on his current CGM reading of 110 mg/dL, the app suggests a precise insulin dose before he starts eating.

Scenario sheet given to the user: You are having dinner at a restaurant and have just been served a plate of pasta. Before you start eating, use the app to find out how much insulin you should take, and save the meal.

SCENARIO 2 • PATIENT Tasks 3 & 4

A low on the way home

At 6 p.m. João is on the bus home after the gym. His phone vibrates with an alert: the app predicts his glucose will reach 65 mg/dL in 20 minutes and suggests eating 15 g of fast-acting carbohydrates. He opens the alert, drinks a small pack of juice he always carries, and taps "Register correction". Later that evening, at home, he opens the Diary to check his day. He remembers that at lunch he left part of the rice on the plate, so he ate fewer carbohydrates than the 60 g he registered, which may explain the low in the afternoon. He opens the lunch entry, changes the carbohydrates to 45 g so that the diary shows what he really ate, keeps the 4.0 units he actually took, and saves the changes.

Scenario sheet given to the user: You are on the bus after the gym and your phone warns you that your glucose is dropping. Deal with the warning. Later, at home, you remember that at lunch you only ate 45 g of carbohydrates instead of the 60 g you registered: correct this in the app.

SCENARIO 3 • DOCTOR Task 2

Routine Clinical Check-up

Dr. Joana is preparing for a consultation with João. She opens the platform's professional dashboard to review his data from the past month. She notices frequent hypoglycemic events during the night and decides to adjust his baseline insulin ratio directly within the app's professional settings.

Scenario sheet given to the user: You are João's endocrinologist and he has an appointment with you tomorrow. Find out how often his glucose went too low during the last month, and change his insulin-to-carbohydrate ratio to 1 unit per 12 g.

6. Observation

Evaluation Observations

Usability Problem Found	**Proposed Solution**
Issue 1: Users hesitated when looking for the diary edit button. It was not clear that tapping the entry would open the edit mode.	Add a visible pencil icon ( ) next to each diary entry so the edit action is immediately recognizable.
Issue 2: During the alert scenario, users dismissed the notification but didn't know where to find the suggested carb correction afterwards.	Keep a persistent 'Active Alert' banner at the top of the Home screen until the user logs the correction.

Users generally understood the purpose of the app from the briefing and were able to start each scenario on their own. However, in the diary scenario, several users hesitated: they looked for an explicit 'edit' button and did not expect that tapping a diary entry would open it for editing. In the alert scenario, some users closed the notification quickly and afterwards were not sure where to find the suggested carbohydrate correction again.